

Pratyusava Baral

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EDUCATION

University of Wisconsin-Milwaukee | *PhD Candidate* 2021 - 2026 (expected)
Thesis title: Detecting and Measuring Gravitational Waves in Current and Future Observatories
Advisors: Jolien Creighton, Patrick Brady

Presidency University, Kolkata | *Bachelor and Masters in Science* 2015 - 2020

FELLOWSHIPS AND HONORS

Wisconsin Space Grant Consortium
Graduate and Professional Research Fellowship 2023, 2024 & 2025

University of Wisconsin-Milwaukee
Distinguished Dissertator Fellowship 2025
Distinguished Graduate Student Fellowship 2023
Chancellor's Graduate Student Award 2021, 2022, 2023, 2024 & 2025
David Lichtman Research Fellowship 2021

PROJECTS

Measuring Speed of Gravity using xG Detectors
Developed a Fisher formulation for estimating errors on measured speed of gravity

Best skymap selector
Implemented a neural network-based selector capable of selecting the best skymap for multimessenger followup from detection pipeline outputs using PyTorch, achieving a 2% reduction in searched area compared to traditional methods

Bilby-xG
Implemented Earth's rotation, detector size effects and multibanding and heterodyne likelihoods capable of handling higher-order modes in Bilby

GstLAL
Contributed to the development of offline and online analysis

RESEARCH INTERESTS

Gravitational-wave data analysis; machine learning for astrophysical inference; Bayesian parameter estimation; next-generation detector science

SOFTWARE DEVELOPMENT & TECHNICAL SKILLS

Programming Languages
Python (10 years) - Primary language for scientific computing, and data analysis
SQL (4 years) - Manage large databases produced by gravitational-wave detection pipelines
C/C++ (10 years)

Software Packages & Libraries

Machine Learning: PyTorch, JAX

Scientific Computing: NumPy, SciPy, pandas, matplotlib

Gravitational Wave: Bilby, LALSuite, GstLAL

High-Performance Computing: HTCondor

Version Control: Git/GitHub

SELECT PUBLICATIONS

During my PhD, I published three first-author papers marked by asterisks (**).

1. ****Baral, P.**, Messick, C. & Brady, P. A Neural Network Approach to Preferred Event Selection for Low-Latency Gravitational-Wave Alerts. arXiv: 2511.14997 [astro-ph.IM] (2025).
2. ****Baral, P.**, Morisaki, S., Gupta, I. & Creighton, J. Parameter estimation of gravitational-wave signals with frequency-dependent antenna responses and higher modes. *Classical and Quantum Gravity* **42**, 215007. <https://doi.org/10.1088/1361-6382/ae128b> (2025).
3. Essick, R., Coughlin, M. W., Zevin, M., Chatterjee, D., Clarke, T. A., Colloms, S., Mali, U., Miller, S., Steinle, N., **Baral P.**, *et al.* Compact binary coalescence sensitivity estimates with injection campaigns during the LIGO-Virgo-KAGRA Collaborations' fourth observing run. *Phys. Rev. D* **112**, 102001. <https://link.aps.org/doi/10.1103/44x3-hv3y> (2025).
4. Joshi, P., Tsukada, L., Hanna, C., Adhicary, S., Mukherjee, D., Niu, W., Sakon, S., Singh, D., **Baral P.**, *et al.* New Methods for Offline GstLAL Analyses. arXiv: 2506.06497 [gr-qc] (June 2025).
5. Niu W., Hanna C., Haster C. J., Adhicary S, **Baral P.** *et al.* GW231109_235456: A Sub-threshold Binary Neutron Star Merger in the LIGO-Virgo-KAGRA O4a Observing Run? arXiv: 2509.09741 [astro-ph.HE] (2025).
6. Chaudhary, S. S., Toivonen, A., Waratkar, G., Mo, G., Chatterjee, D., Antier, S., Brockill, P., Coughlin, M. W., Essick, R., Ghosh, S., Morisaki, S., **Baral P.**, *et al.* Low-latency gravitational wave alert products and their performance at the time of the fourth LIGO-Virgo-KAGRA observing run. *Proceedings of the National Academy of Sciences* **121**, e2316474121. eprint: <https://www.pnas.org/doi/pdf/10.1073/pnas.2316474121> (2024).
7. Ewing, B., Huxford, R., Singh, D., Tsukada, L., Hanna, C., Huang, Y.-J., Joshi, P., Li, A. K. Y., Magee, R., Messick, C., Pace, A., Ray, A., Sachdev, S., Sakon, S., Tapia, R., Adhicary, S., **Baral, P.**, *et al.* Performance of the low-latency GstLAL inspiral search towards LIGO, Virgo, and KAGRA's fourth observing run. *Phys. Rev. D* **109**, 042008. <https://link.aps.org/doi/10.1103/PhysRevD.109.042008> (2024).
8. ****Baral, P.**, Morisaki, S., Hernandez, I. M. & Creighton, J. Localization of binary neutron star mergers with a single cosmic explorer. *Phys. Rev. D* **108**, 043010. <https://link.aps.org/doi/10.1103/PhysRevD.108.043010> (2023).
9. **Baral P.**, Ray, A., Koley, R. & Majumdar, P. Gravitational waves with orbital angular momentum. *The European Physical Journal C* **80**, 326. arXiv: 1901.08804 [gr-qc] (2020).

10. **Baral, P.**, Roy, S. K. & Pal, S. Prospects of probing dark energy with eLISA: Standard versus null diagnostics. *Monthly Notices of the Royal Astronomical Society* **500**, 2896–2907. <https://doi.org/10.1093/mnras/staa3346> (2020).

As a member of the LIGO-Virgo-KAGRA collaboration since 2021, I have co-authored several papers. Highlighted below are the ones in which I have made a direct contribution.

11. The LIGO Scientific Collaboration, the Virgo Collaboration & the KAGRA Collaboration. GW231123: a Binary Black Hole Merger with Total Mass $190\text{--}265\ M_{\odot}$. arXiv: 2507.08219 [astro-ph.HE] (2025).
12. The LIGO Scientific Collaboration, the Virgo Collaboration & the KAGRA Collaboration. GW241011 and GW241110: Exploring Binary Formation and Fundamental Physics with Asymmetric, High-spin Black Hole Coalescences. *The Astrophysical Journal Letters* **993**, L21. <https://doi.org/10.3847/2041-8213/ae0d54> (2025).
13. The LIGO Scientific Collaboration, the Virgo Collaboration & the KAGRA Collaboration. GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog. arXiv: 2508.18080 [gr-qc] (2025).
14. The LIGO Scientific Collaboration, the Virgo Collaboration & the KAGRA Collaboration. GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients. arXiv: 2508.18081 [gr-qc]. <https://arxiv.org/abs/2508.18081> (2025).
15. The LIGO Scientific Collaboration, the Virgo Collaboration & the KAGRA Collaboration. GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run. arXiv: 2508.18082 [gr-qc] (2025).

CONTRIBUTED PRESENTATIONS

1. **LIGO-Virgo-KAGRA Meeting**, Fort Collins, CO Sept 2025
The Age-old Question: What Event Should We Send? Or How I Learned to Stop Worrying and Trust the Machine
2. **xG Mock Data Challenge**, State College, PA June 2025
Measuring Modified Dispersion Relation in the era of Next Generation Observatories
3. **LIGO-Virgo-KAGRA Meeting**, Melbourne, Australia Mar 2025
Neural Network-Based Preferred Event Selector
4. **CosmoMingle 2**, Kolkata, India Jan 2025
Measuring Speed of Gravity and Dispersion using GW370817 in A Single Cosmic Explorer
5. **Hearing beyond the standard model with cosmic sources of Gravitational Waves**, Bangalore, India Dec 2024
Measuring Speed of Gravity and Modified Dispersion Relation in the era of Next Generation Observatories
6. **34th Annual Wisconsin Space Conference**, Milwaukee, WI Aug 2024
Parameter Estimation of Gravitational Wave Signals with Frequency-dependent Antenna Responses and Higher-modes
7. **xG Mock Data Challenge 2024**, State College, PA May 2024
Localization of Binary Neutron Star Mergers using one Cosmic Explorer

8. **CosmoMingle 1**, Kolkata, India Jan 2024
Localization of Gravitational Wave Sources using one Cosmic Explorer
9. **Institute for Cosmic Ray Research**, Kashiwa, Japan Sept 2023
Localization of Binary Neutron Star Mergers using one Cosmic Explorer using Multibanding
10. **33rd Annual Wisconsin Space Conference**, Madison, WI Aug 2023
Bayesian Parameter Estimation of Long Signals in Next Generation Ground-Based Observatories
11. **Midwest Relativity Meeting**, Rochester, MI Oct 2022
Localization of Binary Neutron Star mergers using a Single Cosmic Explorer
12. **Polish Society on Relativity**, Warsaw, Poland *Presented Virtually* Dec 2020
Measuring Angular Momentum from Gravitational Waves
13. **Presision Presidency Physics Summit**, Kolkata, India Aug 2019
Gravitational Waves carrying Orbital Angular Momentum
Best Talk Award

TEACHING EXPERIENCE

Graduate Teaching Assistant, University of Wisconsin-Milwaukee

Physics in Everyday Life	Fall 2024
Calculus-based Mechanics Lab	Spring 2022, 2023 & Fall 2024
Undergraduate Quantum Mechanics	Spring 2021

SERVICE AND OUTREACH

University of Wisconsin-Milwaukee

Organizer, SGN-calibration Meeting	2025
Organizer, GstLAL Face-to-face Meeting	2024
Member, CoffeeShop Astrophysics	2021 - 2025
Founder and Treasurer, Colorlit, a student organization	2023 - 2025
Member, Science and Technology Studies Working Group,	
Center for 21 st Century Studies	2023 - 2025

Vigyan Samagam, Department of Science and Technology, Government of India

Volunteer for SKA, Kolkata chapter	2020 - 2021
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Presidency University, Kolkata

Founder, Presision	2016
Class Representative	2016 - 2017